



**REPUBLIC OF KENYA**

**ASSESSMENT REQUIREMENTS**

**FOR**

**INFORMATION AND COMMUNICATION TECHNOLOGY OPERATOR**

**LEVEL 4**

**PROGRAM CODE - 0612354A**



**TVET CDACC  
P.O. BOX 15745-00100  
NAIROBI**

First published 2025

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## **FOREWORD**

The provision of quality education and training is fundamental to the Government's overall strategy for social economic development. Quality education and training will contribute to achievement of Kenya's development blueprint and sustainable development goals.

Reforms in the education sector are necessary for the achievement of Kenya Vision 2030 and meeting the provisions of the Constitution of Kenya 2010. The education sector had to be aligned to the Constitution and this resulted to the formulation of the Policy Framework for Reforming Education and Training (Sessional Paper No. 1 of 2019). A key feature of this policy is the radical change in the design and delivery of the TVET training. This policy document requires that training in TVET be competency based, curriculum development be industry led, certification be based on demonstration of competence and mode of delivery allows for multiple entry and exit in TVET programmes.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this curriculum has been developed.

It is my conviction that this curriculum will play a great role towards development of competent human resource for the ICT sector's growth and sustainable development.

**PRINCIPAL SECRETARY, VOCATIONAL AND TECHNICAL TRAINING  
MINISTRY OF EDUCATION**

## **PREFACE**

Kenya Vision 2030 aims to transform the country into a newly industrializing, “middle-income country providing a high-quality life to all its citizens by the year 2030”. Kenya intends to create a globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through life-long education and training. TVET has a responsibility of facilitating the process of inculcating knowledge, skills and attitudes necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency Based Education and Training (CBET).

The Technical and Vocational Education and Training Act No. 29 of 2013 on Reforming Education and Training in Kenya, emphasized the need to reform curriculum development, assessment and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry as well as increase the global competitiveness of Kenyan labour force.

TVET Curriculum Development, Assessment and Certification Council (TVET CDACC) in conjunction with ICT Sector Skill Advisory Committee developed Occupational Standards for Information and Communication Technology Operator. These standards were the basis for development of Assessment Requirements for Information and Communication Technology Operator Level 4.

This Assessment Requirements is designed and organized with an outline of Learning Outcomes, Performance Evidence, Knowledge Evidence and Assessment Conditions. They are to be used alongside the Occupational Standards while undertaking CBET assessment. The compilation of the Portfolio of Evidence (PoE) will be based on the Assessment Requirements.

I am grateful to the Council Members, Council Secretariat, ICT SSAC, expert workers and all those who participated in the development of this Assessment Requirements.

**CHAIRPERSON, TVET CDACC**

## **ACKNOWLEDGEMENTS**

These Assessment Requirements were developed through the combined efforts of various stakeholders from private and public organizations. We wish to acknowledge the invaluable contribution received from the stakeholders who provided inputs towards the development of the occupational standards against which this Assessment Requirements was developed.

We are most sincerely thankful to the heads of these institutions who released their staff to join in this important course. Our gratitude goes to the various facilitators that moderated several workshops and ensured that all deliberations and outputs were captured and compiled. These persons did not only demonstrate patience, but also provided leadership by motivating and guiding the groups towards the finalization of this Assessment Requirements. We cannot forget to thank the government agencies that regulate the Technical and Vocational Education and Training (TVET) system namely TVET Authority and CDACC through whom guidance and support was provided on this curriculum development.

Last but not least, we are grateful to any other person, institution, organization or company who played any role in making this process successful but has not been mentioned. We dearly acknowledge your contribution and support.

**COUNCIL SECRETARY/CEO  
TVET CDACC**

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## **ABBREVIATIONS AND ACRONYMS**

|        |                                                               |
|--------|---------------------------------------------------------------|
| CBET   | : Competency Based Education and Training                     |
| CDACC  | : Curriculum Development Assessment and Certification Council |
| CUR    | : Curriculum                                                  |
| KCSE   | : Kenya Certificate of Secondary Education                    |
| KNQA   | : Kenya National Qualifications Authority                     |
| MoEST  | : Ministry of Education Science and Technology                |
| NOS    | : National Occupation Standard                                |
| OS     | : Occupational Standard                                       |
| OSHA   | : Occupation Safety and Health Act                            |
| PPE    | : Personal Protective Equipment                               |
| RPL    | : Recognition of Prior Learning                               |
| SSAC   | : Sector Skills Advisory Committee                            |
| TVETA  | : Technical and Vocational Education and Training Authority   |
| HTTP   | : Hypertext Transfer Protocol                                 |
| ICT    | : Information Communication Technology                        |
| IT     | : Information Technology                                      |
| TCP/IP | : Transmission Control Protocol/Internet Protocol             |
| TVET   | : Technical Vocational Education and Training                 |
| IEEE   | : Institute of Electrical and Electronics Engineers           |
| KNQF   | : Kenya National Qualification Framework                      |
| ISCED  | : International Standard Classification of Education          |

## **ASSESSMENT REQUIREMENTS FOR PERFORM COMPUTER ESSENTIALS**

The assessment requirements cover three components: performance evidence, knowledge evidence and assessment conditions as described below:

### **PERFORMANCE EVIDENCE**

The candidate must show evidence of the ability to complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the job role.

There must be demonstrated evidence that the candidate has completed the following tasks at least once in line with workplace procedures:

1. Manage computer devices
  - 1.1 Worn PPEs (masks, lab coat, antistatic gloves)
  - 1.2 Selected computer hardware devices
  - 1.3 Disassembled computer hardware devices
  - 1.4 Assembled computer hardware devices
  - 1.5 Performed computer booting process
  - 1.6 Connected computer peripheral devices
2. Manage desktop settings
  - 2.1. Customised desktop icons
  - 2.2. Set desktop date and time
  - 2.3. Configured desktop settings
3. Perform file management
  - 3.1 Created files and folders
  - 3.2 Transferred files and folders between various media
  - 3.3 Performed file protection
4. Manage computer software
  - 4.1 Selected data backup media
  - 4.2 Performed data back up
  - 4.3 Installed computer software
  - 4.4 Optimised computer software
5. Perform online jobs
  - 5.1 Identified online job platforms
  - 5.2 Created online accounts and profiles.
  - 5.3 Identified online jobs
  - 5.4 Managed online digital identity
  - 5.5 Done online job bidding
  - 5.6 Executed online tasks
  - 5.7 Managed personal online payment account



## KNOWLEDGE EVIDENCE

The candidate must be able to demonstrate essential knowledge required to effectively complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the work role. This includes knowledge of:

1. Managing computer devices
  - 1.1 Selection of hardware devices
  - 1.2 Disassembling of computer hardware devices
  - 1.3 Assembling of computer hardware devices
  - 1.4 Booting a computer
  - 1.5 Connecting computer peripheral devices
2. Managing desktop settings
  - 2.1. Customisation of desktop icons
  - 2.2. Date and time settings
  - 2.3. Desktop setting customisation
3. Performing file management
  - 3.1 Creating files and folders
  - 3.2 Transferring files and folders
  - 3.3 File protection
4. Managing computer software
  - 4.1 Selecting data backup media
  - 4.2 Performing data backup
  - 4.3 Installation of computer software
  - 4.4 Optimization of computer software
5. Performing online jobs
  - 5.1 Types of online jobs
  - 5.2 Online job platforms
  - 5.3 Online account and profile management
  - 5.4 Online jobs bidding
  - 5.5 Online digital identity
  - 5.6 Executing online tasks
  - 5.7 Management of online payment accounts

## ASSESSMENT CONDITIONS

1. Skills must be demonstrated working individually on the job or in a workplace simulated facility with relevant process, equipment, materials, work instructions and deadlines to enable assessment of candidate's skills and knowledge.
2. Assessment resources required must include:

### **Tools and equipment**

- Computers (desktop/laptop)
- Computer repair toolkit
- System Software suite

- Application software
- External storage (flash disk/external hard drive)
- Uninterruptible Power Supply (UPS)
- Internet connection
- Printer

**Materials and supplies**

- Personal protective Equipment (PPEs)
  - Installation manual
  - Installation CDS /DVDS
  - Printing papers
  - Stapler and staples
3. Simulated assessment environments must simulate the real working environment where these skills and knowledge would be performed with all the relevant equipment and resources.
  4. Assessors must meet the following requirements:
    - 4.1 For those training in a TVET institution they must be:
      - 4.1.1 Licensed by TVET Authority as trainers.
      - 4.1.2 Registered by TVET CDACC as assessors/verifiers in this unit of competency.
    - 4.2 For industry experts they must be registered by TVET CDACC as assessor/verifiers in this unit of competency.

## **ASSESSMENT REQUIREMENTS FOR PERFORM COMPUTER OPERATIONS**

The assessment requirements cover three components: performance evidence, knowledge evidence and assessment conditions as described below:

### **PERFORMANCE EVIDENCE**

The candidate must show evidence of the ability to complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the job role.

There must be demonstrated evidence that the candidate has completed the following tasks at least once in line with workplace procedures:

1. Process computerized word documents
  - 1.1 Created Word documents
  - 1.2 Created and manipulated tables
  - 1.3 Performed mail merging
  - 1.4 Inserted Word processing objects
  - 1.5 Generated list of figures and table of contents
  - 1.6 Printed computerized Word documents
2. Manipulate computerized spreadsheet
  - 2.1. Created Spreadsheet workbook
  - 2.2. Performed cell referencing
  - 2.3. Applied formula and functions
  - 2.4. Generated charts
3. Maintain computerized database
  - 3.1 Collected user requirements for a computerized database
  - 3.2 Designed schemas for computerized database
  - 3.3 Created computerized database objects
  - 3.4 Performed data manipulation
4. Prepare Power point presentation
  - 4.1 Collected PowerPoint Presentation requirements
  - 4.2 Created PowerPoint slides
  - 4.3 Exhibited presentation views
  - 4.4 Performed transitions and animations
  - 4.5 Printed PowerPoint slides

5. Manipulate graphic application
  - 5.1. Identified graphic design requirements
  - 5.2. Created graphic design
  - 5.3. Published graphic design
6. Perform document production
  - 6.1. Printed documents
  - 6.2. Scanned documents
  - 6.3. Duplicated documents
7. Perform Online Collaboration
  - 7.1. Identified online collaboration tools
  - 7.2. Prepared online collaboration
  - 7.3. Applied online collaborative tools
  - 7.4. Demonstrated Mobile collaborations

## **KNOWLEDGE EVIDENCE**

The candidate must be able to demonstrate essential knowledge required to effectively complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the work role. This includes knowledge of:

1. Processing computerized word document
  - 1.1. Ergonomics risk factors
  - 1.2. Creation of computerized Word document
  - 1.3. Creation and manipulation of tables
  - 1.4. Mail merging
  - 1.5. Inserting Word processing objects
  - 1.6. Generating list of figures and table of content
  - 1.7. Printing of computerized Word document
2. Manipulating computerized spreadsheet
  - 2.1. Creation of Computerized spreadsheet workbook
  - 2.2. Cell referencing
  - 2.3. Formulas and functions
  - 2.4. Charts generation
3. Maintaining computerized database
  - 3.1. Computerised database user requirements collection
  - 3.2. Design Computerised database schema
  - 3.3. Creation of Computerised database objects
  - 3.4. Data manipulation
4. Preparing Power point presentation
  - 4.1. Collecting PowerPoint Presentation requirements
  - 4.2. Creating PowerPoint slides
  - 4.3. Exhibition of presentation views

- 4.4. Applying animations and transitions
- 4.5. Manipulation of PowerPoint slides
- 4.6. Printing of PowerPoint slides
5. Manipulating graphic application
  - 5.1. Identifying graphic design requirements
  - 5.2. Creation of graphic design
  - 5.3. Publishing of graphic design
6. Performing document production
  - 6.1. Printing documents
  - 6.2. Document scanning
  - 6.3. Document duplication
7. Performing Online Collaboration
  - 7.1. Identification of Online collaboration tools
  - 7.2. Online collaboration preparation
  - 7.3. Application of online collaborative tools
  - 7.4. Demonstrating Mobile collaborations

## **ASSESSMENT CONDITIONS**

1. Skills must be demonstrated working individually on the job or in a workplace simulated facility with relevant process, equipment, materials, work instructions and deadlines to enable assessment of candidate's skills and knowledge.
2. Assessment resources required must include:

### **Tools and equipment**

- Computer (Desktop / Laptop)
- Application software suite
- Power Supply & Backup (UPS)
- Internet Connection
- Printer
- External Storage Devices (Flash drive / external hard drive)

### **Materials and supplies**

- Printing papers
- Toner/cartridges

3. Simulated assessment environments must simulate the real working environment where these skills and knowledge would be performed with all the relevant equipment and resources.
4. Assessors must meet the following requirements:
  - 4.1 For those training in a TVET institution they must be:
    - 4.1.1 Licensed by TVET Authority as trainers.
    - 4.1.2 Registered by TVET CDACC as assessors/verifiers in this unit of competency.

4.2 For industry experts they must be registered by TVET CDACC as assessor/verifiers in this unit of competency.

## **ASSESSMENT REQUIREMENTS FOR SET UP COMPUTER NETWORK**

The assessment requirements cover three components: performance evidence, knowledge evidence and assessment conditions as described below:

### **PERFORMANCE EVIDENCE**

The candidate must show evidence of the ability to complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the job role.

There must be demonstrated evidence that the candidate has completed the following tasks at least once in line with workplace procedures:

1. Terminated computer cables
  - 1.1 Worn PPEs (lab coat, antistatic gloves)
  - 1.2 Selected Network devices
  - 1.3 Performed network cable trunking
  - 1.4 Terminated Network Cable
2. Connected computer network cables
  - 2.1 Observed safety measures
  - 2.2 Set up Network devices
  - 2.3 Tested Network Cables
  - 2.4 Connected Network Cables
  - 2.5 Established Network connection
  - 2.6 Performed Network testing
3. Perform computer network maintenance
  - 3.1 Monitored Computer network
  - 3.2 Troubleshoot Computer network
  - 3.3 Optimized Computer network

### **KNOWLEDGE EVIDENCE**

The candidate must be able to demonstrate essential knowledge required to effectively complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the work role. This includes knowledge of:

1. Terminating computer cables
  - 1.1. Selecting Network devices
  - 1.2. Network cable trunking
  - 1.3. Network cable termination
2. Connecting computer cables and devices
  - 2.1. Observing safety measures in networking
  - 2.2. Setting up network devices
  - 2.3. Network cable testing
  - 2.4. Network cable connection
  - 2.5. Network testing

## 2.6. Network connection establishment

### 3. Performing Computer network maintenance

#### 3.1. Monitoring computer network

#### 3.2. Troubleshooting Computer network

## **ASSESSMENT CONDITIONS**

1. Skills must be demonstrated working individually on the job or in a workplace simulated facility with relevant process, equipment, materials, work instructions and deadlines to enable assessment of candidate's skills and knowledge.

2. Assessment resources required must include:

### **Tools and equipment**

- Networking toolkit
- Networking cables
- Network devices
- Computers (desktop/laptop)

### **Materials and supplies**

- Manuals
- Personal protective Equipment (PPEs)
- RJ45
- Antistatic gloves

3. Simulated assessment environments must simulate the real working environment where these skills and knowledge would be performed with all the relevant equipment and resources.
4. Assessors must meet the following requirements:
  - 4.1 For those training in a TVET institution they must be:
    - 4.1.1 Licensed by TVET Authority as trainers.
    - 4.1.2 Registered by TVET CDACC as assessors/verifiers in this unit of competency.
  - 4.2 For industry experts they must be registered by TVET CDACC as assessor/verifiers in this unit of competency.



## **ASSESSMENT REQUIREMENTS FOR PERFORM COMPUTER REPAIR AND MAINTENANCE**

The assessment requirements cover three components: performance evidence, knowledge evidence and assessment conditions as described below:

### **PERFORMANCE EVIDENCE**

The candidate must show evidence of the ability to complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the job role.

There must be demonstrated evidence that the candidate has completed the following tasks at least once in line with workplace procedures:

1. Performed computer troubleshooting
  - 1.1 Assessed user data
  - 1.2 Identified computer problems
  - 1.3 Determined solutions to computer problems
2. Repaired faulty components
  - 2.1 Selected Computer components for replacement
  - 2.2 Assembled Tools for repairing or replacing
  - 2.3 Observed safety procedures
  - 2.4 Repaired or replaced faulty computer components
  - 2.5 Disposed obsolete or faulty computer components
3. Test computer component functionality
  - 3.1 Switched on Computer for POST test.
  - 3.2 Tested Computer components
  - 3.3 Generated Computer component's functionality report
4. Perform computer maintenance
  - 4.1 Scheduled Computer maintenance.
  - 4.2 Performed Computer maintenance
  - 4.3 Generated Computer maintenance report

### **KNOWLEDGE EVIDENCE**

The candidate must be able to demonstrate essential knowledge required to effectively complete tasks outlined in elements and performance criteria of this unit, manage tasks and manage contingencies in the context of the work role. This includes knowledge of:

1. Performing computer troubleshooting
  - 1.1. User data assessment
  - 1.2. Computer problems identification

- 1.3. Determining solution to the problem
2. Repairing faulty components.
  - 2.1. Selection of computer components for replacement
  - 2.2. Assembly of tools for repairing or replacing
  - 2.3. Observation of Safety procedures
  - 2.4. Repair and replacing computer components
  - 2.5. Disposing faulty or obsolete computer hardware components
3. Testing computer component functionality
  - 3.1. Performing POST on computer
  - 3.2. Performing computer component test
  - 3.3. Computer component's functionality report
4. Performing computer maintenance
  - 4.1. Computer maintenance scheduling
  - 4.2. Performing computer maintenance
  - 4.3. Computer maintenance report

## **ASSESSMENT CONDITIONS**

1. Skills must be demonstrated working individually on the job or in a workplace simulated facility with relevant process, equipment, materials, work instructions and deadlines to enable assessment of candidate's skills and knowledge.
2. Assessment resources required must include:

### **Tools and equipment**

- Computer repair toolkit
- Computer (desktop/laptop)
- External hard drive
- Flash disk
- Computer hardware diagnostic tools (POST card, multimeter)
- Cleaning tools
- Computer laboratory
- Software tools

### **Materials and supplies**

- Personal protective Equipment (PPEs)
- Installation manual
- Storage facilities

3. Simulated assessment environments must simulate the real working environment where these skills and knowledge would be performed with all the relevant equipment and resources.
4. Assessors must meet the following requirements:
  - 4.1 For those training in a TVET institution they must be:
    - 4.1.1 Licensed by TVET Authority as trainers.

- 4.1.2 Registered by TVET CDACC as assessors/verifiers in this unit of competency.
- 4.2 For industry experts they must be registered by TVET CDACC as assessor/verifiers in this unit of competency.